Observational Incompleteness: From Deterministic Dynamics to Quantum Observations

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Abstract

Observational incompleteness (OI) is the hypothesis that every physical observer is a proper part of a finite reversible deterministic system and records only the visible marginals of its dynamics. At every finite horizon, three classes of law on visible trajectories coincide: all finite stochastic laws, the visible marginals of finite reversible dynamics, and the fixed-basis Born records of finite unitary systems. The components of this correspondence are known, from the trajectory-labelling realization of Egri et al. and Barandes's treatment of permutation dynamics as unistochastic processes; it is stated here as one theorem. Each class contains the Markov laws, so representability does not distinguish a law with memory from one without, and the Born exponent is not selected. The discriminating content concerns memory: every deterministic completion carries the visible memory in its hidden state, and on a fixed finite reversible system finite recurrence turns history readback into stochastic indivisibility somewhere in the recurrence cycle (each with one step not machine-checked). Exact finite operational quantum mechanics is characterized by five completion conditions, none implied by OI alone. Open are which of conditions (iii)–(v) a physical substratum supplies, the selection of the evolution relating two times, and any test of the hypothesis. Applied to the cosmological constant, OI places the zero-point sum and the effective cosmological constant at different levels of description; whether the zero-point sum sources curvature depends on covariant field equations that are not derived, so this is a proposal, not a resolution. Applied to black-hole information, finite computations separate the conservation of information under bijective dynamics from its recovery by an observer with

restricted access; the framework's Page curve is not established, and the information problem is not resolved.

1. Introduction: Observational Incompleteness and the emergence of quantum behaviour

1.1 Embedded observation

Embedded observation is modelled by a triple (S, φ, V) [Main §1.2]: a finite total system $S = \mathcal{C}_V \times \mathcal{C}_H$ of observer configurations $\mathcal{C}_V$ and complement configurations $\mathcal{C}_H$, a bijection φ of S, and an observer V that registers $x \in \mathcal{C}_V$ and not $h \in \mathcal{C}_H$.

The bijection φ is the reversible representative of the hidden dynamics, motivated by observed near-unitarity, and finiteness of all of S is posited [Main §1.2, Remark (status of the early commitments)]. Hidden states compatible with the same x send it to different futures, so the visible sector evolves stochastically although the whole evolves deterministically.

Observational incompleteness (OI) is the physical hypothesis that every physical observer is in this situation. Identifying nature with such a system is a representation choice, motivated empirically; the theorems below are mathematical, and no test of the identification at the level of memory witnesses is established [Main §4.5]. The incompleteness is inaccessibility of the hidden sector, not joint unsharpness of non-commuting observables: it is present for a single fixed readout, and no uncertainty relation constrains fixed-readout visible laws [Main §3.4].

1.2 The idea and its two consequences

Marginalizing a reversible dynamics over an unobserved register produces stochastic laws, conditional memory, and failures of stochastic divisibility, in which no stochastic intermediate transition map exists (where the candidate intermediate matrix exists, it has negative entries). The stochastic–quantum correspondence relates these features to quantum theory [Bar23; Bar25]. At the level of finite-horizon observable laws the correspondence of §2 (Theorem 1) is exact and universal, so the content lies in where the memory of a law must sit (§2.4) and what finite recurrence does with it (§2.5). The paper develops one idea, OI: its foundation and quantum-emergence results, which stop short of operational quantum mechanics (§§2–3), and two of its consequences, for the cosmological constant (§4) and for black-hole information (§5). Neither application is a demonstrated resolution of its problem; §6 states what would decide each. Proofs, the status ledger and the probe code are in [Main], which this paper introduces; gravitational applications are in [GR] and [Book].

Contributions. The paper claims no new theorem or physical result; its contribution is expository. It presents the established mathematical results of OI together, separates the mathematical correspondence from its physical interpretation, and states two consequences with their conditions and limitations.

The black-hole results of §5 are reported from unpublished repository research records [BHa; BHb], with written proofs and exact or exhaustive checks, whose novelty has not been established. The components of Theorem 1 are known [EGGH26; Bar23; Bar25; Bar26; SGD19]; the paper states them as one theorem with each direction attributed. The memory bounds of §2.4 are forms of the data-processing and Pinsker inequalities. The recurrence argument of §2.5 combines finite order with the known contraction criterion for divisibility [RHP14; BD16], stated concurrently by Doukas [Dou26]. The results of §3.2–§3.3 are proved in [GR] and [Main].

Conventions. "Machine-checked" means checked in Lean 4/Mathlib [dMU21; Mathlib20] at commit `17b830bc` by the recipe under "Code, data and formal verification" (declarations in Appendix A); "proved on paper" means a written proof without a machine check; an "exact probe" is an exhaustive computation in exact rational arithmetic at small size. A *faithful realization* of a law is a bijection φ of $\mathcal{C}_V \times \mathcal{C}_H$ with an initial law under which the visible trajectory has that law; the finite reversible system (S, φ) is the *substratum*.

2. Theoretical foundation: the established results and their assumptions

2.1 Setting

Fix finite sets $\mathcal{C}_V$ (with $|\mathcal{C}_V| \geq 2$) and $\mathcal{C}_H$, a bijection φ of $\mathcal{C}_V \times \mathcal{C}_H$, and a horizon K. A law is a probability distribution on visible trajectories $(x_0, \dots, x_K) \in \mathcal{C}_V^{K+1}$.

The visible law of a realization depends on the dynamics, the partition and the initial law; the prior is part of the realization datum, not selected by invariance under φ, and the counting measure is a maximal-entropy default [Main §1.2, Remark (the measure as a realization datum)].

Three classes of law are compared [Main §3.4]:

- S_K: all probability distributions on $\mathcal{C}_V^{K+1}$.
- D_K: the visible trajectory laws of a bijection of $\mathcal{C}_V \times \mathcal{C}_H$, for some finite $\mathcal{C}_H$, under an initial law that may correlate the visible and hidden coordinates.
- Q_K: the laws of finite-dimensional unitary systems read out by projective Born measurement in one fixed basis at every step, from an initial state diagonal in that basis, the record being a fixed function of each outcome.

2.2 The finite-horizon correspondence

Theorem 1 (finite-horizon stochastic–reversible–unitary equivalence [Main §3.4]). *For every finite visible alphabet and every finite horizon K,*

$$\mathsf{S}_K = \mathsf{D}_K = \mathsf{Q}_K,$$

so each class is the set of all probability laws on $\mathcal{C}_V^{K+1}$.

Theorem 1 is machine-checked as `finite_horizon_equivalence`, with the inclusions $\mathsf{S}_K \subseteq \mathsf{D}_K$ (`S_imp_D`), $\mathsf{D}_K \subseteq \mathsf{Q}_K$ (`D_imp_Qfb`) and $\mathsf{Q}_K \subseteq \mathsf{S}_K$ (`Qfb_imp_S`), and the reverse inclusions as their compositions; no construction enters as a hypothesis.

Realization. For $\mathsf{S}_K \subseteq \mathsf{D}_K$ the realization stores a sampled trajectory, with a clock, in the hidden register [Main §3.4, Theorem (finite-horizon stochastic–reversible–unitary equivalence); EGGH26, Prop. 13]. For a law with rational probabilities, a realization whose hidden prior is independent of x_0 also exists [Main §3.4, Theorem (finite-horizon process dilation)]; it is proved on paper, with exact probes.

Quantum representation. For $\mathsf{D}_K \subseteq \mathsf{Q}_K$ the representation is the permutation unitary $U_\varphi|x,h\rangle = |\varphi(x,h)\rangle$, read out in the configuration basis at every step [Main §3.4]; $\mathsf{Q}_K \subseteq \mathsf{S}_K$ is the Born rule.

Attribution. The trajectory-labelling realization is that of Egri et al. [EGGH26, Prop. 13], packaged as one iterated bijection; the independent-prior form is their Prop. 16, without reversibility, and its Markov one-step case is the random-mapping representation [LP17, Prop. 1.5]. A permutation dynamics is a special case of a unistochastic process [Bar25, p. 12; Bar23 §4.3; Bar26 §4], and the one-step n^2-dimensional ancilla dilation of [Main §3.1] coincides with the constructions of Barandes [Bar26, Thm 3.3] and Schmidt [Sch21, Thm 2]. The every-step projectively measured record as a probability law is in [SGD19, Eq. (3)]; that work and [Milz20] study whether such records are consistent with omitting measurements, a question Q_K, which measures at every step, does not meet.

2.3 Scope of the correspondence

Markov laws lie in all three classes. So neither a finite reversible realization nor a fixed-basis unitary representation distinguishes a law with memory from one without [Main §3.4, Remark (the conditions are diagnostics, not hypotheses)].

The relation $p = |U|^2$ is part of the definition of Q_K, not an output. The unitaries supplied are permutations, with entries 0 and 1, so the recovered table is the same for $|U|^p$ at every $p > 0$. The quadratic exponent is therefore not selected [Main §3.4; Bar26, Thm 3.1].

A finite bijection has finite order L, so one fixed realization has an L-periodic multi-time law: the correspondence holds per horizon, and a nonperiodic law has no exact all-time realization by one fixed finite substratum [Main §3.4].

2.4 Where the memory lives

Write $M_t := I(X_{<t}; X_{t+1} \mid X_t)$ for the conditional memory at time t. In a realization, write $f_x(h) = \pi_V(\varphi(x,h))$, with π_V the projection onto $\mathcal{C}_V$, and μ_p for the hidden posterior after a visible history p ending at x.

Theorem 2 (unavoidable hidden predictive memory [Main §3.4]). *In every faithful realization:*

- *(a)* $P(X_{t+1} \mid p, x) = (f_x)_{\#}\mu_p$*;*
- *(b) for histories* p, p' *ending at* x*,* $\|P(X_{t+1} \mid p, x) - P(X_{t+1} \mid p', x)\|_{\mathrm{TV}} \leq \|\mu_p - \mu_{p'}\|_{\mathrm{TV}}$*;*
- *(c)* $M_t \leq I(X_{<t}; H_t \mid X_t) \leq \log_2 |\mathcal{C}_H|$.

(a)–(c) are machine-checked for a joint law of history, present and hidden state. Their identification with a realization's time-t law is not machine-checked. The written proof is in [Main §3.4, Theorem (unavoidable hidden predictive memory)]; see also [CT06].

Corollary (proved on paper [Main §1.5]). $M_t > 0$ holds exactly when every faithful realization carries readout-relevant hidden memory at t, that is, a pair of posteriors with different pushforwards. The forward direction rests on Theorem 2(a) in each realization; the converse on Theorem 2(a) in one realization, which Theorem 1 supplies (`S_imp_D`); both use the standard vanishing criterion for conditional mutual information [CT06], which, like the time-t identification, is not machine-checked.

Two further bounds tie memory to the hidden sector. A realization of a law whose conditional past–future information reaches I^* needs $|\mathcal{C}_H| \geq 2^{I^*}$ (machine-checked in the same form as Theorem 2). If two values of $X_{<k}$ followed by a common present state X_k, each of joint probability at least p_0, induce next-step laws at total-variation distance at least δ, then $I(X_{<k}; X_{k+1} \mid X_k) \geq p_0\delta^2/\ln 2$ (in bits), by Pinsker's inequality (proved on paper, with exact probes) [Main §2.3; CT06].

[Main §1.3] names four conditions on a hidden sector: (C1) influence, (C2) persistence of records, (C3) capacity, and (C4) history readback, in which two histories with the same present induce different next-step laws. In a faithful realization (C4) coincides with $M_t > 0$ by the vanishing criterion [CT06], and the Corollary places it in the hidden state; (C1) and (C3) follow from it, and (C2) is a physical-regime premise. The conditions are diagnostics of a realization, not hypotheses of Theorem 1.

2.5 Recurrence and indivisibility

Write Γ_t for the *rooted* visible transition matrix from time 0 to time t (rows indexed by x_0). The family is *P-divisible on* $\{0, \dots, n\}$ if for all $s < t \leq n$ some stochastic Λ satisfies $\Gamma_t = \Gamma_s\Lambda$. A stochastic divisor contracts total variation, so under P-divisibility $\|\delta_a\Gamma_t - \delta_b\Gamma_t\|_{\mathrm{TV}}$ cannot increase. This is the divisibility criterion of [RHP14, Thm 2.4] and the data-processing direction of [BD16, Prop. 1].

Theorem 3 (recurrence and overlap [Main §2.3]). *Let* φ *be a bijection of* $\mathcal{C}_V \times \mathcal{C}_H$ *with a fixed hidden prior. If two rows of* Γ_s *overlap in a column and* $\Gamma_n = I$ *for some* $n > s$*, the family is not P-divisible on* $\{0, \dots, n\}$.

Theorem 3 is machine-checked in its rooted form, and Doukas states the same overlap-and-return argument concurrently [Dou26, Prop. 1]. Monotone distance is necessary for P-divisibility but not sufficient [Main §2.3].

A finite square stochastic matrix with a stochastic one-sided inverse is a permutation matrix [BB12, Problem 23] (machine-checked). With this lemma, on a fixed finite reversible substratum, history readback forces indivisibility somewhere in the recurrence cycle [Main §2.3, Theorem (history readback + finite recurrence implies indivisibility)]. The conclusion is machine-checked, by the route of Theorem 3, from a routed-readback form of (C4), in which a visible-history record written into the hidden state is read back later [Main §1.3, the realization clause of (C4)]; from history-level (C4) it is proved on paper.

Appendix B works examples: a minimal model of Theorem 3 with six hidden states, memory without short-window indivisibility, and a two-card game, the smallest realization of Theorem 3.

3. Quantum emergence: what is demonstrated, and what remains open for equivalence with quantum mechanics

3.1 What is demonstrated

The indivisibility of §2.5 is global: it occurs somewhere in the recurrence cycle, not on every window. Conditional memory does not force short-window indivisibility (Appendix B).

Three notions are distinct, and no unconditional implication holds among them [Main §3.2, Remark (three notions, kept apart)]: (1) accessible memory, $M_t > 0$; (2) stochastic P-indivisibility of the visible population law; (3) quantum non-Markovianity under a chosen criterion. Tracing out a permutation unitary from a diagonal hidden state gives channels that preserve diagonal states, and for them (2) implies CP-indivisibility [Main §3.2, Theorem (CP-indivisibility, framework scope)]. Outside that class the implication fails: with Φ_{t_1} conjugation by the Hadamard gate and $\Phi_{t_2} = \mathrm{id}$ the family is CP-divisible, while its computational-basis populations, $\frac{1}{2}J \to I$ with J the all-ones matrix, are P-indivisible [Main §3.2, Remark (scope: P-indivisibility is not quantum non-Markovianity)]. Distinguishability backflow witnesses divisibility failure in one direction only [BLP09]; for bijective dynamical maps on a d-dimensional system, CP-divisibility is equivalent to monotone distinguishability with a $(d+1)$-dimensional ancilla [BJA17].

Theorem 1 is a representation result: it states which laws admit realizations and quantum representations, not why a physical system exhibits a law with memory [Main §3.4, Remark (representation, not explanation)]. Calvo states a proof of the six axioms of textbook quantum mechanics from the single axiom that every physical system evolves according to a generally indivisible stochastic law [Cal26]; Theorem 1 is a correspondence between classes of law and derives no such axioms.

3.2 From the correspondence to operational quantum mechanics

Operational quantum mechanics requires, beyond Theorem 1's fixed-basis statistics, coherent preparations, noncommuting interventions and composites represented by one instrument theory.

Fix a nonempty finite system and an operational theory specifying which finite outcome families of linear maps are available on the system and at every ancilla level. The theory contains the identity and is closed under coarse-graining of outcomes and, at every ancilla level, under outcome-conditioned composition; it has an available preparation of a uniformly mixed ancilla, which an available operation may follow, an available spectator-independent local readout of the ancilla, and ancilla discard: an available preparation followed by an available instrument, with the ancilla then discarded, is an available instrument on the system with the same outcomes [GR §3.3]. The five conditions are: (i) available families send positive states to positive states and their outcome traces sum to the input trace; (ii) operations remain available with a one-state ancilla adjoined; (iii) an untouched independent system can be adjoined without changing an intervention; (iv) every finite unitary on every composite is available; (v) attaching a fresh uniformly prepared ancilla to a composite, acting, and discarding it gives an available operation.

Theorem 4 (operational-completion characterization [GR §3.3]). *For every nonempty finite system, the available outcome families on the system and at every positive ancilla level are exactly the normalized finite Kraus instruments if and only if conditions (i)–(v) hold.*

Both directions are machine-checked, $\Rightarrow$ as `physical_of_exactAll` and $\Leftarrow$ as `exactAll_of_physical_general`, for every nonempty finite carrier.

On the qubit, each condition is independent of the other four and of a fixed finite observation process satisfying (C1)–(C4). For each condition there is a qubit theory that realizes that process and the other four conditions, and fails it. In particular, a qubit theory that never creates coherence realizes the observation process and is not quantum. So OI alone does not imply quantum mechanics. Which of (iii)–(v) a physical substratum supplies is open [GR §3.3].

3.3 The evolution relating two times is not selected

Representability does not fix the unitary evolution relating two times [Main §3.4, Remark (scope: representability, not selection of the relating evolution)]. Two coherent lifts of one visible family, unitary dilations that reproduce the visible law at every time and compose exactly across times, can differ by a time-dependent right action invisible to the visible law and then have different relating evolutions, while a constant deformation cancels from every relating evolution (both machine-checked).

Take a lift with trivial ancilla at one time, and form the fixed-basis datum whose evolution is the lift's unitary, with uniform initial law and identity

readout. If two such data have the same one-step visible matrix, their root-conditioned trajectory laws agree at every horizon (`bridge_traj`); conversely, if their rooted families agree, their one-step visible matrices are equal (`rooted_eq_iff_slice_eq`, forward direction). Each such trajectory law is fixed-basis realizable and stochastic (`realData_traj_stochastic`). The fixed-basis correspondence therefore acts on the quotient by visible equality. A selection condition beyond representability and ordinary coherence would be needed, and none is supplied here.

3.4 No fixed finite carrier reproduces a qubit

A fixed finite realization with N ontic configurations generates only behaviour matrices that factor nonnegatively through N states. Simulating a qubit's prepare-and-measure statistics needs classical memory unbounded over finite experiments: explicit tasks are given by Galvão and Hardy [GH03] and Montina [Mon08], and Montina reports Hardy's result that the number of ontic states cannot be finite [Mon08; Har04]. So no fixed finite realization reproduces the full prepare-and-measure theory of a qubit exactly; the factorization is proved, and the unboundedness is imported [Main §3.4, Proposition (finite-substratum operational obstruction — classical-dimension form); HKLP24, Thm 4]. The obstruction does not touch Theorem 1, whose realization is chosen per law.

3.5 The Born exponent and Bell structure

The origin of the quadratic Born exponent, which the correspondence does not select (§2.3), is open [Main §3.4; Bar26, Thm 3.1].

A deterministic completion that keeps pointwise causal cones and a setting-independent pre-setting ensemble is Bell-local, $|S_{\mathrm{CHSH}}| \leq 2$ [Bel64; CHSH69; Main §3.3], so a deterministic completion reproducing quantum Bell correlations must give up one of the two. The framework keeps measurement independence and adopts ontic parameter dependence, in which one wing's microstate response depends on the remote setting without operational signalling, and the compatibility of that choice with operational no-signalling and the gravitational continuum limit of the substratum's spatial coupling graph is the open hypothesis H-Bell [Main §3.3].

4. The cosmological constant: how OI might change the interpretation of vacuum energy

Summing zero-point energies of the quantum fields up to a Planck-scale cutoff gives a vacuum energy density some 120 orders of magnitude above the observational value [Wei89, §I; §III, Eqs. (3.4)–(3.6)] (the mismatch), the exact power depending on the cutoff convention; if that density sources curvature, it must be cancelled to that accuracy [Wei89, §III].

In OI (§1.1) the zero-point sum of the emergent quantum description (§§2–3)

and the effective cosmological constant are defined at different levels of description. A uniform shift of the emergent Hamiltonian changes no visible transition probability, so the offset has no content in the visible statistics of the quantum description [GR §6.2]. The invariance is the familiar statement that no non-gravitational measurement sees the absolute vacuum energy; OI's proposal is that the mismatch requires no cancellation, and its content lies in how the two levels couple to geometry.

Whether the offset also fails to source curvature is a separate coupling question. Settling it requires covariant field equations derived from the substratum, with their coupling to sources (Level G3 of [GR]), which are open [GR §6.2]. The quantum description therefore does not decide between gravitational completions that differ by a term $\rho_C\, g_{\mu\nu}$, such as $\mathcal{G}_{\mu\nu} = 8\pi G\, T^{\mathrm{cl}}_{\mu\nu}$ and $\mathcal{G}_{\mu\nu} = 8\pi G\,(T^{\mathrm{cl}}_{\mu\nu} - \rho_C\, g_{\mu\nu})$, with $\mathcal{G}_{\mu\nu}$ the gravitational side of the field equations (the Einstein tensor in general relativity), $T^{\mathrm{cl}}_{\mu\nu}$ the classical stress-energy of the substratum and ρ_C the energy density of the offset [GR §6.2]. The proposal that the mismatch requires no cancellation is therefore conditional on that open step.

If the covariant field equations take the form [GR] targets, $E_{\mu\nu}[g, \mathrm{OI}] = 8\pi G\,(T_{\mu\nu} - \mathcal{F}[g,T]\, g_{\mu\nu})$, with $E_{\mu\nu}[g, \mathrm{OI}]$ the gravitational side still to be derived, $\mathcal{F}[g,T]$ a scalar functional of the metric and the stress-energy, and the classical stress-energy of the substratum as source, the zero-point sum of the emergent description is not a source and needs no cancellation; the vacuum term that remains, from the substratum's classical energy density and from $\mathcal{F}$, is not computed. It does not explain the observed value of Λ, which enters as a cosmological input. The scale $\rho \sim H^2_{\mathrm{Hub}}/G$, with H_{Hub} the Hubble rate, is the Friedmann normalization of the total density, and the fraction carried by Λ is an input.

With a Planck-scale cutoff the mismatch equals the de Sitter entropy up to a numerical factor fixed by the cutoff convention; this is a dimensional identity, not an independent prediction. How the energy of electroweak and QCD vacuum transitions couples to gravity is not fixed by the descriptive distinction and requires the same open coupling step.

Removing the vacuum energy of matter as a source of curvature is also the aim of the trace-free Einstein equations, which leave the observed value of Λ unexplained [EvEMU11], and of vacuum-energy sequestering, which also renders contributions from phase transitions small [KP14]; Volovik argues from the thermodynamics of quantum many-body ground states that the ultraviolet zero-point sum does not set the vacuum energy [Vol06]. The present proposal differs in locating the gravitational source in the substratum microstate, and its coupling step is open.

A derivation of the covariant field equations, with their conservation and source coupling, from the substratum would turn the proposal into a theorem or refute it. Until then the proposal reinterprets the problem and does not resolve it.

5. Black-hole information: preservation versus accessible recovery

The black-hole information problem asks whether the radiation of an evaporating black hole carries the information of what formed it [Haw76; Pag93; Pag13]. Because the dynamics of OI is a bijection observed through a partition (§1.1), two questions separate: whether information is conserved in the global reversible dynamics, and whether it can be recovered from the records an observer can access, its *access class*. The results below are finite computations in toy models of configurations, permutations and configuration reads. They concern the framework's nested-partition construction [GR App. A; Book §7.6], not a particular black hole.

Under OI, information is conserved in the global reversible dynamics because the dynamics is a bijection, which is a premise of the framework. Conservation neither implies nor is necessary for recovery from the records an observer can access: finite exhaustive computations exhibit bijections with no accessible advantage, that is, no gain in success probability over a blind guess, and non-injective maps with exact recovery [BHa].

Let $\mathcal{U}$ be the set of joint values of the registers left unread at the end and $\mathcal{M}$ the message set. Of the 40,320 bijections of $\{0,1\}^3$, 1,376 give zero accessible advantage for a binary message with one register unread after two emissions. For bijective dynamics, complete endpoint access gives exact recovery, and with $\mathcal{U}$ inaccessible $P_{\text{succ}} \geq 1/|\mathcal{U}|$, so zero advantage in a bijective instance requires $|\mathcal{U}| \geq |\mathcal{M}|$; 32 non-injective maps of $\{0,1\}^2$ violate the bound on P_{succ} [BHa]. These endpoint guarantees are sufficient, not necessary. The bound is the remnant-capacity form of a known fact, and its linear case is the secret-sharing threshold [BHa; Sha79].

The Page-curve calculation of [GR App. A] addresses recovery under its own conditions and is a conditional result [Book §7.6, §18.5]. Its configuration space is $\mathcal{C}_V \times \mathcal{C}_B \times \mathcal{C}_D$, with V the exterior visible region, B the black-hole interior and D the sector beyond the cosmological horizon; the radiation R, a subsystem of V, is emitted from B [GR App. A.1, A.7]. For all but an exponentially negligible fraction of initial conditions, the full-cycle time average of the radiation's entanglement entropy, at the bipartition of each stage held fixed, equals the Page value $S_{\text{Page}}(d_B, d_R)$ for factor dimensions d_B, d_R, up to the correction term of Theorem A.8 [Pag93; GR, Thms A.8–A.9]. This holds under the scrambling hypothesis H-scramble, that the orbit ensemble equidistributes over quantum pure states on the shell; H-scramble fails for every bijection acting on a definite initial configuration with the bipartition aligned to the configuration coordinates and is not established [GR App. A.7; BHb, C1, C6(a)–(b)]. It also needs the decoupling condition H-decouple: $B \cup R$ evolves with no coupling to D or to the rest of the visible sector, $V \setminus R$, and its joint state is pure, over whole cycles of the dynamics; timescale ordering does not supply it [BHa, item F], and it is not established [GR App. A.7]. Theorem A.8 of [GR] takes the whole

configuration space $\mathcal{C}_B \times \mathcal{C}_R$ as a single energy shell. That requires the energy to be constant there, so that energy conservation places no constraint on the orbit, and it does not follow from the conservation of a Hamiltonian that takes more than one value there. An evaporating black hole whose radiation carries away its energy is in the proper-shell case, where the mean entropy depends on the shell and in general differs from the Page value; that case is open [GR App. A.7; BHb, C6(c)]. The result is a time average at a fixed bipartition, and whether the entropy along a single trajectory follows the same curve is open [GR App. A.7; BHb, C6(d)]. In the dependency ladder of [GR] the Page curve is also conditional on the covariant field equations. The generalized second law is not derived [GR App. A.6]; agreement with quantum-extremal-surface computations [Pen20] is not established; the correspondence between islands and substratum cycles is conjectural [Book §7.6]. The Page curve of [GR App. A] is therefore not established, and the black-hole information problem is not resolved here.

6. Discussion and conclusion: implications, limitations and decisive open tests

6.1 Implications

If the hypothesis holds, the fixed-basis statistics of quantum theory are the visible laws of an embedded observer, and their departures from Markov behaviour are memory carried in an inaccessible sector; whether the rest of operational quantum mechanics follows depends on completion conditions that OI does not imply (§3.2). Quantum Bell correlations are not accounted for by memory alone: with measurement independence kept, they need ontic parameter dependence, whose compatibility with operational no-signalling and the continuum limit is the open hypothesis H-Bell (§3.5).

6.2 What is not claimed

1. No derivation of quantum mechanics: Theorem 1 contains every finite law, and the Born exponent is not selected.
2. No unique selection of the operational theory: conditions (i)–(v) are not implied by OI alone, their independence is shown on the qubit, which of (iii)–(v) a physical substratum supplies is open, and the visible law does not fix the relating evolution.
3. No characterization of memory by the existence of a realization (§2.3), no identification of P-indivisibility with quantum non-Markovianity, and no claim that memory forces indivisibility on short windows (§3.1).
4. No resolution of the black-hole information problem: the Page curve of [GR App. A] is not established (§5), and the results of §5 reported from [BHa; BHb] concern finite toy models, each proved on paper or checked exhaustively at small size or by exact arithmetic. No claim is made that information reaches D, that [GR, Thm A.7] fails within its stated scope,

or that no other decoupling argument exists.

5. No claim that the physical hypothesis is established or tested.
6. No claim that the card game (Appendix B) produces quantum rather than classical statistics: on card inputs its statistics are classical stochastic ones; under the unitary embedding its one-step channel is entanglement-breaking and its two-step family admits a classical-memory decomposition; it violates no Bell inequality.
7. No prediction of the value of Λ or Ω_Λ, and no resolution of the cosmological constant problem: §4 is conditional on the covariant field equations.
8. No claim that the viewpoint is new: deterministic systems analysed with quantum methods have been studied before, for example in [tH14]. What the paper states are the cited theorems and their scopes.

6.3 Decisive open tests

For quantum emergence: whether a physical substratum supplies completion conditions (iii)–(v), and whether history readback (C4) holds at the cosmological partition [GR §2.2]; no experiment distinguishing OI from standard quantum theory, and no memory-witness constraint, is established [Main §4.5]. For the cosmological constant: a derivation of the covariant field equations with their source coupling. For black-hole information: whether H-scramble and H-decouple hold, the Page value on a proper energy shell and along a single trajectory, and, in [GR]'s ladder, the covariant field equations.

Further open problems are: (1) machine checks of the steps of §2.2, §2.4 and §2.5 that lack one; (2) whether the minimal hidden predictive memory equals M_t once ledger bits that cannot influence the future are quotiented (the fixed-dimension form is false) [Main §3.4]; (3) exact gluing for recurrent target families, and approximation on windows short compared with the recurrence time [Main §3.4]; (4) (C2) necessity, which is conditional on a hidden-sector mixing hypothesis motivated by eigenstate thermalization [Main §3.4].

6.4 Conclusion

At every finite horizon the laws an embedded observer of a finite reversible system can record are exactly the finite stochastic laws, and exactly the fixed-basis unitary Born laws; because the representation is universal, the discriminating content lies in memory, which every deterministic completion carries in its hidden state. Exact operational quantum mechanics is reached only under completion conditions that OI alone does not imply, and representability does not select the evolution relating two times.

Applied to vacuum energy, OI proposes that the zero-point sum of the emergent description is not a gravitational source, conditional on covariant field equations that are not derived; applied to black-hole information, it separates conservation from accessible recovery, while the framework's Page curve is not established. Neither application is a demonstrated resolution of its problem.

Code, data and formal verification

The declarations in Appendix A were checked in Lean 4.33.0 with Mathlib v4.33.0 at commit 17b830bcd0ecc36e10eff5b9eeca03bdd7198d86, whose Lean sources are identical at the corpus commit `b64919cd`, by building the Lean project in `verification/lean-mathlib` (`lake exe cache get`, `lake --rehash build`) and running `python3 tools/release_gate.py`, whose `lean-axioms` step requires every `#print axioms` report to lie within `propext`, `Classical.choice` and `Quot.sound`. Each declaration in Appendix A reports exactly these three; the expected result and the comparison recipe are in `papers/introduction_repro/`.

The probes and the black-hole computations [BHa; BHb] are released under the MIT License, at these repository paths at the corpus commit:

- `papers/oi_lattice_code/foundations/` (probes; also in the release archive, concept DOI 10.5281/zenodo.19060318);
- `verification/programmes/black-hole-information/` (black-hole computations; not part of the release archive).

Appendix A. Machine-checked declarations

Modules of the `OIBridge` Lean library; line numbers at commit `17b830bc`.

- **Theorem 1** (§2.2): `finite_horizon_equivalence` (Equivalence:462); directions `S_imp_D` (:422), `D_imp_Qfb` (:294), `Qfb_imp_S` (:245), compositions :466–468; supporting `permMatrix_mem_unitaryGroup` (EquivalenceChain:179). Prior independence is proved on paper.
- **Theorem 2** (§2.4): `pushforward_identity`, `distinguishability_floor`, `capacity_floor`, `unavoidable_hidden_predictive_memory` (HiddenMemory:83, :114, :170, :185), for a joint law of history, present and hidden state; the time-t identification and the standard vanishing criterion for conditional mutual information [CT06] are not machine-checked.
- **Capacity bound** (§2.4): `c3_necessity`, `card_hidden_ge_two_pow_Istar` (C3Necessity:80, :117).
- **Stochastic-inverse lemma; Theorem 3** (§2.5): `isPermMatrix_of_stochastic_inverse` (EquivalenceChain:110); `pIndivisible_of_rooted_identity_return_of_overlap` (RecurrenceHorizon:148).
- **Readback forces indivisibility** (§2.5): `routed_forces_indivisible_somewhere` (PhysicalC4Discharge:122) for routed readback; on paper for (C4).
- **Relating evolution** (§3.3): `gl2_strong_gauge_moves_relative_candidate`, `gl3_constant_gauge_preserves_relative` (CoherentLiftGauge:537, :451); `bridge_traj`, `rooted_eq_iff_slice_eq` (forward direction), `realData_traj_stochastic` (TrackBQfbBridge:276, :180, :286).
- **Theorem 4** (§3.2): `exactAll_iff_physical_general` (GeneralCarrier:100); ⇒ `physical_of_exactAll` (PhysicalCharacterization:301); ⇐ `exactAll_of_physical_general` (GeneralCarrier:93) via `krausSoundExt_of_validity_inert` (OperationalValidity:124),

`compositeCompleteness_unconditional` (IsometryExtension:179, premise discharged by `finiteIsometryExtensionSF_discharged`, :116) and `exactAll_of_levelOne` (LevelOneSeam:196).

- **Independence; OI alone does not imply QM** (§3.2, qubit): `five_way_minimality` (RankGapTheory:702), `oi_alone_not_qm` (GeneralCarrier:160).

Appendix B. Examples

B.1 A minimal model

Let $\mathcal{C}_V = \{0,1\}$ and $\mathcal{C}_H = \{1,\dots,6\}$ [Main §2.4]. For $h \in \{1,2\}$, φ swaps $(0,h)$ with $(1,h)$. It flips the visible bit and keeps h, which then records the flip. At fixed x, φ swaps $h=3$ with 4, and 5 with 6. A second application reads the record and flips the bit back. With h uniform:

$$\Gamma_1 = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}, \qquad \Gamma_2 = I, \qquad \Gamma_1^{-1}\Gamma_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The only candidate divisor has negative entries, so the family is P-indivisible. A Markov chain with the same one-step matrix would instead give Γ_1^2, with diagonal entries $5/9$ and off-diagonal entries $4/9$.

With x_0 uniform and independent of h, $P(x_2{=}0 \mid x_1{=}1, x_0{=}0) = 1$ while $P(x_2{=}0 \mid x_1{=}1) = 1/3$. For $P(x_0{=}0) = a$, the second value is $a/(2-a)$.

B.2 Memory without short-window indivisibility

Let X_0 and X_1 be independent uniform bits and $X_2 = X_0 \oplus X_1$. Every two-time transition matrix is $\frac{1}{2}J$, so the family factors through $\Lambda = I$ and is P-divisible. Yet $I(X_0; X_2 \mid X_1) = 1$ bit. Up to flipping the final bit, this is the process of [RHP14, Eq. (15)]; it appears concurrently in [Dou26, App. A].

B.3 The card table

The two-card game of [Book §1.11] uses two cards, each red or black: a visible *hole card* h, and a face-down *burn card* b, uniform and independent of h. A *street* is one shuffle step, in which the two cards trade places: $\varphi(h,b) = (b,h)$. A *peek* reveals the hole card and overwrites the burn card, which holds the record, with a fresh card drawn uniformly and independently.

The probability that the hole card at showdown is the hole card as dealt is obtained by exact enumeration of the equally likely cases:

Protocol	Probability
one shuffle	1/2
two shuffles	1
two shuffles, with a peek between them	1/2
Markov prediction for two shuffles, Γ_1^2	1/2

Here $\Gamma_1 = \frac{1}{2}J$ and $\Gamma_2 = I$. The two rows of Γ_1 are equal, so for every Λ the two rows of $\Gamma_1\Lambda$ are equal and $\Gamma_1\Lambda \neq I$: the family is P-indivisible.

The peek row depends on the rule: if the peek only reveals or replaces the hole card, the probability is 1, so the $1/2$ comes from overwriting the record, not from looking. With anticorrelated cards ($b = 1 - h$), Γ_1 is a permutation and the game is divisible.

On card inputs the game is a classical stochastic process, and its P-indivisibility is notion (2) of §3.1. Embed the game as a permutation unitary acting on an arbitrary hole-card state ρ, with the burn card in the uniform diagonal state [Main §3.2]. The one-step channel is then the constant channel $\rho \mapsto \mathrm{tr}(\rho)\, I/2$, which is entanglement-breaking [Main §3.2, Remark (failure of (C1))]. The two-step map is the identity channel. The family is therefore CP-indivisible [Main §3.2, Theorem (CP-indivisibility, framework scope)].

Following Bäcker, Beyer and Strunz [BBS24], a two-step process has classical memory when the second map can be written as an instrument realizing the first map, followed by channels conditioned on its outcome. With the hidden slot initialized to a fixed card instead, the swap has the maps $\rho \mapsto |0\rangle\langle 0|$ and $\rho \mapsto \rho$ and admits no such decomposition [Main §4.5]. With the uniform burn card it does: the instrument $\rho \mapsto \frac{1}{4}\sigma_o \rho \sigma_o$, $o \in \{0, x, y, z\}$, sums to the one-step map by the identity $\sum_o \sigma_o X \sigma_o = 2\,\mathrm{tr}(X)\, I$, and the channels $\rho \mapsto \sigma_o \rho \sigma_o$, conditioned on o, return ρ.

A card arrangement violates no Bell inequality [Book §1.11; Main §3.3].

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